# VIS-0 Template Check & Vision Feasibility -- Page 1 / 3

Labels: Template check VALID (deterministic); vision replies VALID as outputs (temperature 0); timings and memory NOT measurements (agent resident); quality grading pending (Local AI)

## Step A Verdict: CONFIRMED

CONFIRMED: Gemma 3 markers (<start_of_turn>, <end_of_turn>) are tokenized as multiple plain-text pieces, whereas Gemma 4 markers (<|turn>, <turn|>) are single special tokens and are used by the GGUF chat_template.

### 1. /tokenize with "with_pieces": true:

- '<start_of_turn>user' -> several plain-text pieces (8 pieces) [236820:'<', 3041:'start', 236779:'_', 1340:'of', 236779:'_', 887:'turn', 236813:'>', 2364:'user']
- '<end_of_turn>' -> several plain-text pieces (7 pieces) [236820:'<', 643:'end', 236779:'_', 1340:'of', 236779:'_', 887:'turn', 236813:'>']
- '<|turn>user' -> ONE special token (<|turn>) + role token (2 tokens total) [105:'<|turn>', 2364:'user']
- '<turn|>' -> ONE special token [106:'<turn|>']
- '<|turn>model' -> ONE special token (<|turn>) + role token (2 tokens total) [105:'<|turn>', 4368:'model']

### 2. /apply-template with [{"role": "user", "content": "Hi"}] (verbatim repr):

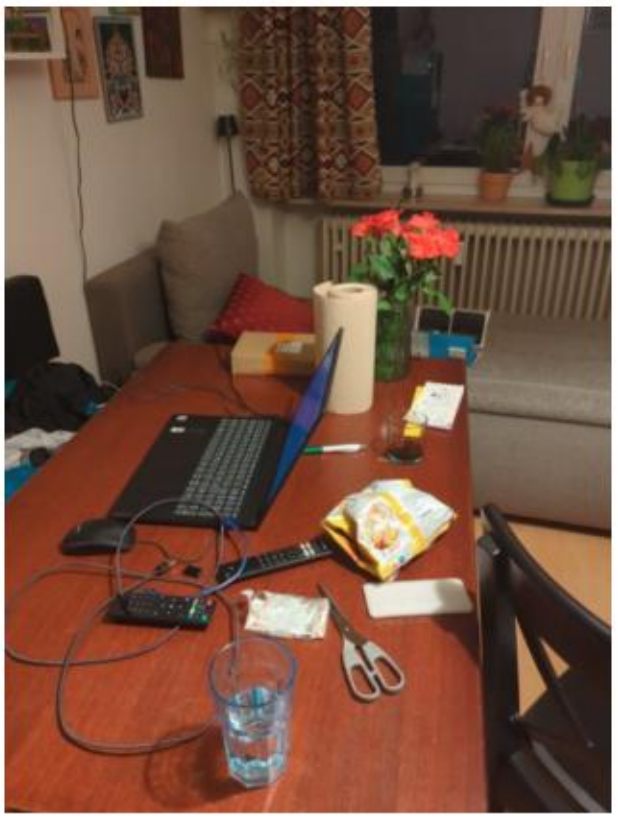
'<|turn>user\nHi<turn|>\n<|turn>model\n'

### 3. /props chat_template header & turn-token lines:

```
{#
  Template: Google Gemma 4 Canonical Chat Template
  Author: Google Gemma Engineering Team
  Published: 2026-07-09
  Context: Fixed tool-calling loops, turn closures, and thinking content-ordering.
#}
...
{{- '<|turn>system\n' -}}
{{- '<turn|>\n' -}}
{{- '<|turn>' + role + '\n' }}
{{- '<turn|>\n' -}}
{{- '<|turn>model\n' -}}
```

### 4. Vision Budgets Summary (NOT measurements; agent resident):

- Budget 70: 2/2 ok, crashes=0, logged_image_tokens=[63]
- Budget 140: 2/2 ok, crashes=0, logged_image_tokens=[154]
- Budget 280: 2/2 ok, crashes=0, logged_image_tokens=[300]
- Optional 140+YOLO: 2/2 ok, crashes=0



YOLO detections (Y-ONLY input lines):
- cup [0], center, far; - chair [1], right, medium; - laptop [2], center, close; - potted plant [3], center, medium; - scissors [4], center, far; - potted plant [5], right, far
- potted plant [6], right, far; - cup [7], center, far; - cup [8], center, far; - couch [9], left, medium; - book [10], center, far; - cell phone [11], left, far; - chair [12], left, medium; - chair [13], left, far; - remote [14], center, far; - vase [15], center, far; - dining table [16], center, very close

Text OLD (/completion, Gemma 3 markers) [prompt_tok=231]:
The dining table is very close in front of the robot.

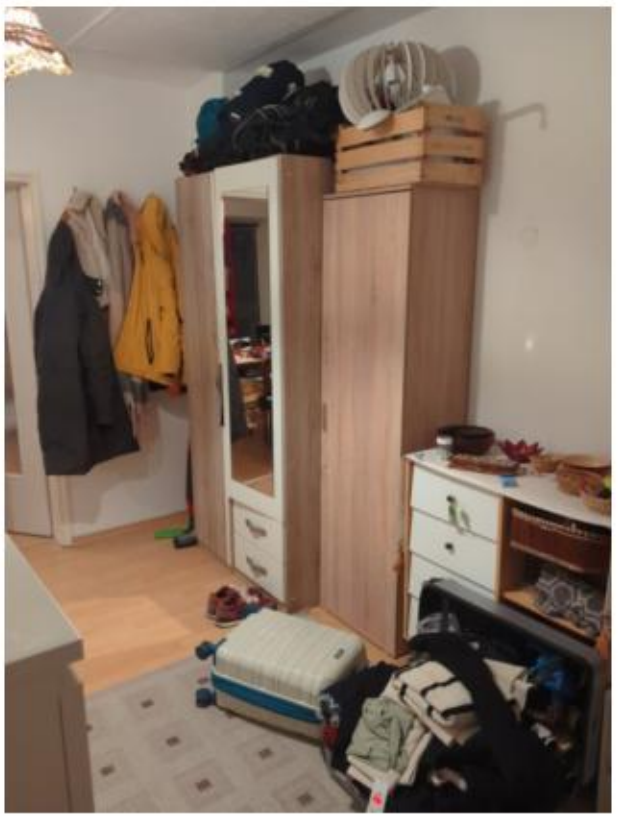
Text NEW (/chat/completions, GGUF template) [prompt_tok=213]:
In front of the robot is a dining table very close, a laptop close, and a potted plant in the center, medium.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a wooden table with a laptop, a blue object, and some other items like a glass and scissors.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a wooden table with various items, including a laptop, a blue and black keyboard, a glass of water, and some scissors.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot, there is a wooden table with a laptop, various cables, a blue and black keyboard, a mouse, a glass of water, and some kitchen items like a knife and a cutting board.

Optional Vision 140 + YOLO [prompt_tokens=359, image_tokens_logged=154]:
In front of the robot is a wooden dining table with a laptop, various cups, a vase, and some potted plants. There are also chairs and a couch visible around the table.



YOLO detections (Y-ONLY input lines):
- bowl [0], right, far; - suitcase [1], right, medium

Text OLD (/completion, Gemma 3 markers) [prompt_tok=83]:
'The robot is looking at the bowl.\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>user\nThe bowl is in front of the robot.\n<start_of_turn>user\nThe bowl is in front of the robot.\n<start_of_turn>user\nThe bowl is in front of the robot.\n<start_of_turn>user\nThe bowl is'

Text NEW (/chat/completions, GGUF template) [prompt_tok=65]:
In front of the robot is a bowl to the right and far, and a suitcase to the right and medium.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a collection of items, including a white and blue patterned object, and a black object.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a collection of items, including a white and black object that looks like a piece of equipment or a bag, and a pile of dark clothing.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot, there is a light-colored rug on the floor and a stack of folded clothes.

Optional Vision 140 + YOLO [prompt_tokens=211, image_tokens_logged=154]:
In front of the robot, there is a white suitcase and a pile of dark clothing.